KCET 2020 Paper

Subject: **Mathematics**
Code: **C4**

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1. If $y = 2x^{n+1} + \frac{3}{x^n}$, then $x^2 \frac{d^2y}{dx^2}$ is
   a. y
   b. $6n(n+1)y$
   c. $n(n+1)y$
   d. $x\frac{dy}{dx} + y$

2. If the curves $2x = y^2$ and $2xy = K$ intersect perpendicularly, then the value of K^2 is
   a. 8
   b. 4
   c. $2\sqrt{2}$
   d. 2

3. If $(xe)^y = e^x$, then $\frac{dy}{dx}$ is
   a. $\frac{e^x}{x(y-1)}$
   b. $\frac{\log x}{(1+\log x)^2}$
   c. $\frac{1}{(1+\log x)^2}$
   d. $\frac{\log x}{(1+\log x)}$

4. If the side of a cube is increased by 5%, then the surface area of a cube is increased by
   a. 20%
   b. 10%
   c. 60%
   d. 6%

5. The value of $\int \frac{1+x^4}{1+x^6} dx$ is
   a. $\tan^{-1}x + \frac{1}{3}\tan^{-1}x^2 + C$
   b. $\tan^{-1}x + \tan^{-1}x^3 + C$
   c. $\tan^{-1}x + \frac{1}{3}\tan^{-1}x^3 + C$
   d. $\tan^{-1}x - \frac{1}{3}\tan^{-1}x^3 + C$

6. The maximum value of $\frac{\log_e x}{x}$, if $x > 0$ is
   a. $-\frac{1}{e}$
   b. e
   c. 1
   d. $\frac{1}{e}$

7. The value of $\int e^{\sin x} \sin 2x \, dx$ is
   a. $2e^{\sin x}(\cos x - 1) + C$
   b. $2e^{\sin x}(\sin x - 1) + C$
   c. $2e^{\sin x}(\sin x + 1) + C$
   d. $2e^{\sin x}(\cos x + 1) + C$

8. The value of $\int_{-\frac{1}{2}}^{\frac{1}{2}} \cos^{-1}x \, dx$ is
   a. $\frac{\pi^2}{2}$
   b. π
   c. $\frac{\pi}{2}$
   d. 1

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9. If $\int \frac{3x+1}{(x-1)(x-2)(x-3)}dx = A\log|x-1| + B\log|x-2| + C\log|x-3| + C$, then the values of A, B and C are respectively,
   a. $2, -7, 5$
   b. $5, -7, -5$
   c. $2, -7, -5$
   d. $5, -7, 5$

10. The value of $\int_0^1 \frac{\log(1+x)}{1+x^2}dx$ is
   a. $\frac{\pi}{8}\log 2$
   b. $\frac{\pi}{2}\log 2$
   c. $\frac{\pi}{4}\log 2$
   d. $\frac{1}{2}$

11. The area of the region bounded by the curve $y^2 = 8x$ and the line $y = 2x$ is
   a. $\frac{8}{3}$ sq. units
   b. $\frac{16}{3}$ sq. units
   c. $\frac{4}{3}$ sq. units
   d. $\frac{3}{4}$ sq. units

12. The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos x}{1+e^x}dx$ is
   a. -2
   b. 2
   c. 0
   d. 1

13. The order of the differential equation obtained by eliminating arbitrary constants in the family of curves $c_1 y = (c_2 + c_3)e^{x+c_4}$ is
   a. 4
   b. 1
   c. 2
   d. 3

14. The general solution of the differential equation $x^2dy - 2xydx = x^4\cos x\, dx$ is
   a. $y = \cos x + cx^2$
   b. $y = x^2\sin x + cx^2$
   c. $y = x^2\sin x + c$
   d. $y = \sin x + cx^2$

15. The area of the region bounded by the line $y = 2x + 1, x -$ axis and the ordinates $x = -1$ and $x = 1$ is
   a. 5
   b. $\frac{9}{4}$
   c. 2
   d. $\frac{5}{2}$

16. The two vectors $\hat{\imath} + \hat{\jmath} + \hat{k}$ and $\hat{\imath} + 3\hat{\jmath} + 5\hat{k}$ represent the two sides $\overrightarrow{AB}$ and $\overrightarrow{AC}$ respectively of a ΔABC. The length of the median through A is
   a. $\sqrt{14}$
   b. $\frac{\sqrt{14}}{2}$
   c. 14
   d. 7

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17. If $\vec{a}$ and $\vec{b}$ are unit vectors and θ is the angle between $\vec{a}$ and $\vec{b}$, then $\sin\frac{\theta}{2}$ is

a. $|\vec{a}-\vec{b}|$
b. $|\vec{a}+\vec{b}|$
c. $\frac{|\vec{a}+\vec{b}|}{2}$
d. $\frac{|\vec{a}-\vec{b}|}{2}$

18. The curve passing through the point $(1, 2)$ given that the slope of the tangent at any point (x, y) is $\frac{2x}{y}$ represents

a. Hyperbola
b. Circle
c. Parabola
d. Ellipse

19. If $|\vec{a}\times\vec{b}|^2 + |\vec{a}\cdot\vec{b}|^2 = 144$ and $|\vec{a}| = 6$, then $|\vec{b}|$ is equal to

a. 4
b. 6
c. 3
d. 2

20. The point $(1, -3, 4)$ lies in the octant

a. Eighth
b. Second
c. Third
d. Fourth

21. If the vector $2\hat{\imath} - 3\hat{\jmath} + 4\hat{k}$, $2\hat{\imath} + \hat{\jmath} - \hat{k}$ and $\lambda\hat{\imath} - \hat{\jmath} + 2\hat{k}$ are coplanar, then the value of λ is

a. 5
b. 6
c. -5
d. -6

22. The distance of the point $(1, 2, -4)$ from the line $\frac{x-3}{2} = \frac{y-3}{3} = \frac{z+5}{6}$ is

a. $\frac{\sqrt{293}}{49}$
b. $\frac{293}{7}$
c. $\frac{\sqrt{293}}{7}$
d. $\frac{293}{49}$

23. The sine of the angle between the straight line $\frac{x-2}{3} = \frac{3-y}{-4} = \frac{z-4}{5}$ and the plane $2x - 2y + z = 5$ is

a. $\frac{\sqrt{2}}{10}$
b. $\frac{3}{\sqrt{50}}$
c. $\frac{3}{50}$
d. $\frac{4}{5\sqrt{2}}$

24. If a line makes an angle of $\frac{\pi}{3}$ with each of x and $y-$ axis, then the acute angle made by $z-$ axis is

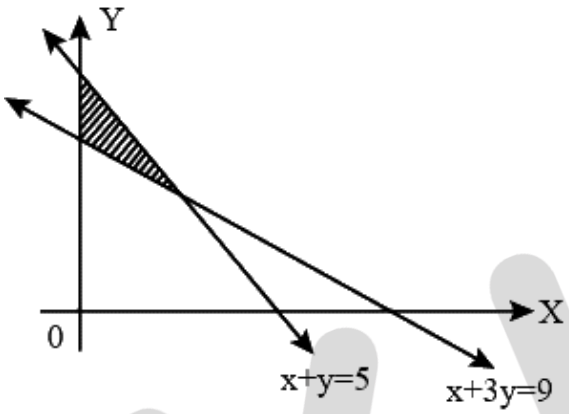
a. $\frac{\pi}{2}$
b. $\frac{\pi}{4}$
c. $\frac{\pi}{6}$
d. $\frac{\pi}{3}$

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25. Corner points of the feasible region determined by the system of linear constraints are $(0, 3), (1, 1)$ and $(3, 0)$. Let $z = px + qy$, where $p, q > 0$. Condition on p and q so that the minimum of z occurs at $(3, 0)$ and $(1, 1)$ is

a. $p = q$
b. $p = 2q$
c. $p = \frac{q}{2}$
d. $p = 3q$

26. The feasible region of an LPP is shown in the figure. If $Z = 11x + 7y$, then the maximum value of Z occurs at



a. $(3, 2)$
b. $(0, 5)$
c. $(3, 3)$
d. $(5, 0)$

27. A die is thrown 10 times, the probability that an odd number will come up at least one time is

a. $\frac{1013}{1024}$
b. $\frac{1}{1024}$
c. $\frac{1023}{1024}$
d. $\frac{11}{1024}$

28. If A and B are two events such that $P(A) = \frac{1}{3}$, $P(B) = \frac{1}{2}$ and $P(A \cap B) = \frac{1}{6}$, then $P(A'|B)$ is

a. $\frac{1}{12}$
b. $\frac{2}{3}$
c. $\frac{1}{3}$
d. $\frac{1}{2}$

29. Events E_1 and E_2 form a partition of the sample space S. A is any event such that $P(E_1) = P(E_2) = \frac{1}{2}$, $P(E_2|A) = \frac{1}{2}$ and $(A|E_2) = \frac{2}{3}$. Then $P(E_1|A)$ is

a. $\frac{1}{4}$
b. $\frac{1}{2}$
c. $\frac{2}{3}$
d. 1

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30. The probability of solving a problem by three persons A, B and C independently is $\frac{1}{2}, \frac{1}{4}$ and $\frac{1}{3}$ respectively. Then the probability that the problem is solved by any two of them is
    a. $\frac{1}{8}$
    b. $\frac{1}{12}$
    c. $\frac{1}{4}$
    d. $\frac{1}{24}$

31. If $n(A) = 2$ and total number of possible relations from set A to set B is 1024, then $n(B)$ is
    a. 5
    b. 512
    c. 20
    d. 10

32. The value of $\sin^2 51° + \sin^2 39°$ is
    a. $\cos 12°$
    b. 1
    c. 0
    d. $\sin 12°$

33. If $\tan A + \cot A = 2$, then the value of $\tan^4 A + \cot^4 A =$
    a. 5
    b. 2
    c. 1
    d. 4

34. If $A = \{1, 2, 3, 4, 5, 6\}$, then the number of subsets of A which contain at least two elements is
    a. 58
    b. 64
    c. 63
    d. 57

35. If $z = x + iy$, then the equation $|z + 1| = |z - 1|$ represents
    a. $y -$ axis
    b. a circle
    c. a parabola
    d. $x -$ axis

36. The value of ${}^{16}C_9 + {}^{16}C_{10} - {}^{16}C_6 - {}^{16}C_7$ is
    a. ${}^{17}C_2$
    b. 0
    c. 1
    d. ${}^{17}C_{10}$

37. The number of terms in the expansion of $(x + y + z)^{10}$ is
    a. 110
    b. 66
    c. 142
    d. 11

38. If $P(n) : 2^n < n!$, then the smallest positive integer for which $P(n)$ is true if
    a. 5
    b. 2
    c. 3
    d. 4

39. The two lines $lx + my = n$ and $l'x + m'y = n'$ are perpendicular if
    a. $lm' + ml' = 0$
    b. $ll' + mm' = 0$
    c. $lm' = ml'$
    d. $lm + l'm' = 0$

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40. If the parabola $x^2 = 4ay$ passes through the point $(2, 1)$, then the length of the latus rectum is
    a. 8
    b. 1
    c. 4
    d. 2

41. If the sum of n terms of an A.P. is given by $S_n = n^2 + n$, then the common difference of the A.P. is
    a. 6
    b. 4
    c. 1
    d. 2

42. The negation of the statement "For all real numbers x and y, $x + y = y + x$" is
    a. for some real numbers x and y, $x - y = y - x$
    b. for all real numbers x and y, $x + y \neq y + x$
    c. for some real numbers x and y, $x + y = y + x$
    d. for some real numbers x and y, $x + y \neq y + x$

43. The standard deviation of the data $6, 7, 8, 9, 10$ is
    a. 10
    b. $\sqrt{2}$
    c. $\sqrt{10}$
    d. 2

44. $\lim_{x \to 0} \left(\frac{\tan x}{\sqrt{2x + 4} - 2} \right)$ is equal to
    a. 6
    b. 2
    c. 3
    d. 4

45. If a relation R on the set $\{1, 2, 3\}$ be defined by $R = \{(1, 1)\}$, then R is
    a. Only symmetric
    b. Reflexive and symmetric
    c. Reflexive and transitive
    d. Symmetric and transitive

46. Let $f : [2, \infty] \to \mathbf{R}$ be the function defined by $f(x) = x^2 - 4x + 5$, then the range of f is
    a. $[5, \infty)$
    b. $(-\infty, \infty)$
    c. $[1, \infty)$
    d. $(1, \infty)$

47. If A, B, C are three mutually exclusive and exhaustive events of an experiment such that $P(A) = 2P(B) = 3P(C)$, then $P(B)$ is equal to
    a. $\frac{4}{11}$
    b. $\frac{1}{11}$
    c. $\frac{2}{11}$
    d. $\frac{3}{11}$

48. The domain of the function defined by $f(x) = \cos^{-1}\sqrt{x - 1}$ is
    a. $[0, 1]$
    b. $[1, 2]$
    c. $[0, 2]$
    d. $[-1, 1]$

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49. The value of $\cos\left(\sin^{-1}\frac{\pi}{3}+\cos^{-1}\frac{\pi}{3}\right)$ is

a. Does not exist
b. 0
c. 1
d. -1

50. If $A=\begin{pmatrix}0&0&1\\0&1&0\\1&0&0\end{pmatrix}$, then A^4 is equal to

a. $4A$
b. A
c. $2A$
d. I

51. If $A=\{a,b,c\}$, then the number of binary operations on A is

a. 3^9
b. 3
c. 3^6
d. 3^3

52. If $\begin{pmatrix}2&1\\3&2\end{pmatrix}A=\begin{pmatrix}1&0\\0&1\end{pmatrix}$, then the matrix A is

a. $\begin{pmatrix}2&-1\\3&2\end{pmatrix}$
b. $\begin{pmatrix}2&1\\3&2\end{pmatrix}$
c. $\begin{pmatrix}2&-1\\-3&2\end{pmatrix}$
d. $\begin{pmatrix}-2&1\\3&-2\end{pmatrix}$

53. If $f(x)=\begin{vmatrix}x^3-x&a+x&b+x\\x-a&x^2-x&c+x\\x-b&x-c&0\end{vmatrix}$, then

a. $f(-1)=0$
b. $f(1)=0$
c. $f(2)=0$
d. $f(0)=0$

54. If A and B are square matrices of same order and B is a skew symmetric matrix, then $A'BA$ is

a. Skew symmetric matrix
b. Symmetric matrix
c. Null matrix
d. Diagonal matrix

55. If A is a square matrix of order 3 and $|A|=5$, then $|A \text{ adj } A|$ is

a. 625
b. 5
c. 125
d. 25

56. If $f(x)=\begin{cases}\frac{1-\cos Kx}{x\sin x}, & \text{if } x\neq 0\\ \frac{1}{2}, & \text{if } x=0\end{cases}$ is continuous at $x=0$, then the value of K is

a. ± 1
b. $\pm\frac{1}{2}$
c. 0
d. ± 2

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57. If $a_1, a_2, a_3, \ldots \ldots, a_9$ are in A.P. then the value of $\begin{vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{vmatrix}$ is

a. 1
b. $\frac{9}{2}(a_1 + a_9)$
c. $a_1 + a_9$
d. $\log_e(\log_e e)$

58. If $2^x + 2^y = 2^{x+y}$, then $\frac{dy}{dx}$ is

a. $\frac{2^y - 1}{2^x - 1}$
b. 2^{y-x}
c. -2^{y-x}
d. 2^{x-y}

59. If $f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$, then $f'(\sqrt{3})$ is

a. $-\frac{1}{\sqrt{3}}$
b. $-\frac{1}{2}$
c. $\frac{1}{2}$
d. $\frac{1}{\sqrt{3}}$

60. The right hand and left limit of the function $f(x) = \begin{cases} \frac{e^{1/x}-1}{e^{1/x}+1}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$ are respectively

a. -1 and 1
b. 1 and 1
c. 1 and -1
d. -1 and -1

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Answer Key

Question Number	1	2	3	4	5	6	7	8	9	10
Correct Answer	(c)	(a)	(b)	(b)	(c)	(d)	(b)	(c)	(a)	(a)

Question Number	11	12	13	14	15	16	17	18	19	20
Correct Answer	(c)	(d)	(b)	(b)	(d)	(a)	(d)	(a)	(d)	(d)

Question Number	21	22	23	24	25	26	27	28	29	30
Correct Answer	(b)	(c)	(a)	(b)	(c)	(a)	(c)	(b)	(b)	(c)

Question Number	31	32	33	34	35	36	37	38	39	40
Correct Answer	(a)	(b)	(b)	(d)	(a)	(b)	(b)	(d)	(b)	(c)

Question Number	41	42	43	44	45	46	47	48	49	50
Correct Answer	(d)	(d)	(b)	(b)	(d)	(c)	(d)	(b)	(a)	(d)

Question Number	51	52	53	54	55	56	57	58	59	60
Correct Answer	(a)	(c)	(d)	(a)	(c)	(a)	(d)	(c)	(c)	(c)

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Solutions

1. (c)

$y = 2x^{n+1} + 3x^{-n}$

$\Rightarrow \frac{dy}{dx} = 2(n+1)x^n - 3nx^{-n-1}$

$\Rightarrow \frac{d^2y}{dx^2} = 2n(n+1)x^{n-1} + 3n(n+1)x^{-n-2}$

$\Rightarrow x^2\frac{d^2y}{dx^2} = n(n+1)\left[2x^{n+1} + \frac{3}{x^n}\right]$

$\Rightarrow x^2\frac{d^2y}{dx^2} = n(n+1)y$

2. (a)

$2x = y^2 \quad \ldots(1)$

$2xy = K \quad \ldots(2)$

Solving (1) and (2), we get

$(x, y) = \left(\frac{K^{\frac{2}{3}}}{2}, K^{\frac{1}{3}}\right)$

Differentiating (1) and (2) w.r.t. x

$m_1 = \frac{dy}{dx} = \frac{1}{y} \quad \ldots(3)$

$m_2 = \frac{dy}{dx} = -\frac{y}{x} \quad \ldots(4)$

$\because$ Both curves intersect each other perpendicularly

$\therefore m_1m_2 = -1 \Rightarrow -\frac{1}{x} = -1 \Rightarrow x = 1$

$\Rightarrow K^{\frac{2}{3}} = 2 \Rightarrow K^2 = 8$

3. (b)

$(xe)^y = e^x \Rightarrow y(\log x + 1) = x$

$\Rightarrow y = \frac{x}{\log x + 1}$

$\therefore \frac{dy}{dx} = \frac{\log x}{(\log x+1)^2}$

4. (b)

Let one side of the cube be x and surface area be A

So, $dx = 5\% = \frac{5x}{100}$

Then, $A = 6x^2$

$\Rightarrow \frac{dA}{dx} = 12x \Rightarrow dA = (12x)dx$

$\Rightarrow dA = 12x \cdot \frac{5x}{100}$

$\Rightarrow dA = \frac{10A}{100} \Rightarrow dA = 10\%$

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5. (c)

$$\int \frac{1+x^4}{1+x^6}dx = \int \frac{1+x^4-x^2+x^2}{(1+x^2)(1-x^2+x^4)}dx$$
$$= \int \left(\frac{1-x^2+x^4}{(1+x^2)(1-x^2+x^4)} + \frac{x^2}{(1+x^2)(1-x^2+x^4)}\right)dx$$
$$= \int \left(\frac{1}{1+x^2} + \frac{1}{3}\cdot\frac{3x^2}{1+(x^3)^2}\right)dx$$
$$= \tan^{-1}x + \frac{1}{3}\tan^{-1}x^3 + C$$

6. (d)

$y = \frac{\log_e x}{x}$

$\Rightarrow \frac{dy}{dx} = \frac{1-\log_e x}{x^2}$

For maxima, $\frac{dy}{dx} = 0$

$\Rightarrow 1 - \log_e x = 0$

$\Rightarrow x = e$

$\frac{dy}{dx}$ changes sign from positive to negative at $x = e$

$\therefore \ y_{\max} = \frac{1}{e}$

7. (b)

$I = \int e^{\sin x} \sin 2x \, dx = 2\int e^{\sin x} \sin x \cos x \, dx$

Let $t = \sin x \Rightarrow dt = \cos x \, dx$

$\therefore I = 2\int te^t dt$

$= 2\,(te^t - e^t) + C = 2(\sin x - 1)e^{\sin x} + C$

8. (c)

$\int_{-\frac{1}{2}}^{\frac{1}{2}} \cos^{-1}x \, dx$

$= |x\cos^{-1}x|_{-\frac{1}{2}}^{\frac{1}{2}} + \int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{x}{\sqrt{1-x^2}}dx$ [Using integration by parts]

$= \frac{1}{2}\cos^{-1}\left(\frac{1}{2}\right) + \frac{1}{2}\cos^{-1}\left(-\frac{1}{2}\right) - \left[\left(\frac{1}{2}\right)\cdot 2\sqrt{1-x^2}\right]_{-\frac{1}{2}}^{\frac{1}{2}}$

$= \frac{1}{2}\cdot\frac{\pi}{3} + \frac{1}{2}\cdot\frac{2\pi}{3} - 0$

$= \frac{\pi}{6} + \frac{\pi}{3} = \frac{\pi}{2}$

9. (a)

Let $\frac{3x+1}{(x-1)(x-2)(x-3)} = \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3}$ $\cdots (1)$

$\Rightarrow \int \frac{3x+1}{(x-1)(x-2)(x-3)}dx = A\log|x-1| + B\log|x-2| + C\log|x-3| + C$

Now, $3x + 1 = A(x-2)(x-3) + B(x-1)(x-3) + C(x-1)(x-2)$ [From eqn.(1)]

Putting $x = 1,\ x = 2,\ x = 3$ in the above equation one at a time, we get

$A = 2,\ B = -7,\ C = 5$

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10. (a)

$I = \int_0^1 \frac{\log(1+x)}{1+x^2} dx$

Let $x = \tan\theta \Rightarrow dx = \sec^2\theta\, d\theta$

$I = \int_0^{\frac{\pi}{4}} \log(1+\tan\theta)\, d\theta$

$\Rightarrow I = \int_0^{\frac{\pi}{4}} \log\left(1+\tan\left(\frac{\pi}{4}-\theta\right)\right) d\theta = \int_0^{\frac{\pi}{4}} \log\left(1+\frac{1-\tan\theta}{1+\tan\theta}\right) d\theta$

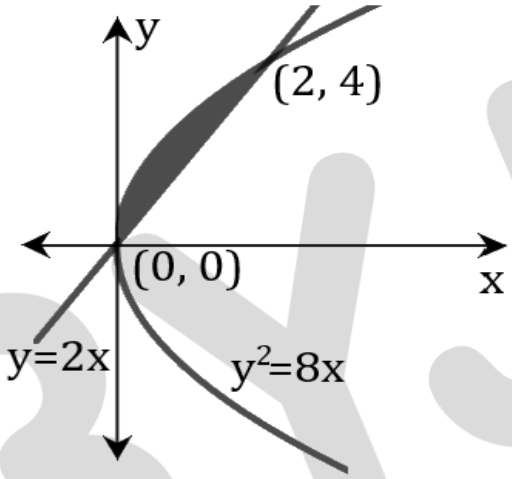
$\Rightarrow I = \int_0^{\frac{\pi}{4}} \log\left(\frac{2}{1+\tan\theta}\right) d\theta = \int_0^{\frac{\pi}{4}} \log 2\, d\theta - \int_0^{\frac{\pi}{4}} \log(1+\tan\theta)\, d\theta$

$\Rightarrow 2I = \frac{\pi}{4}\log 2 \Rightarrow I = \frac{\pi}{8}\log 2$

11. (c)

Solving $y^2 = 8x$ and $y = 2x$, we get

$(x, y) = (0, 0), (2, 4)$



So, area bounded by the curve is

$\int_0^2 \left(2\sqrt{2}x^{1/2} - 2x\right) dx = \left(\frac{4\sqrt{2}}{3}x^{3/2} - x^2\right)_0^2 = \frac{4}{3}$ sq. units

12. (d)

$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos x}{1+e^x} dx \quad \cdots (1)$

$\Rightarrow I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos\left(\frac{\pi}{2}-\frac{\pi}{2}-x\right)}{1+e^{\frac{\pi}{2}-\frac{\pi}{2}-x}} dx$

$\Rightarrow I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos x}{1+e^{-x}}\, dx = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{e^x \cos x}{1+e^x} dx \quad \cdots (2)$

From (1) and (2), we have

$2I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{(e^x+1)\cos x}{e^x+1} dx = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos x\, dx$

$\Rightarrow 2I = \sin x\,|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \Rightarrow I = 1$

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13. (b)

$y = \left[\left(\frac{c_2+c_3}{c_1}\right)e^{c_4}\right]e^x = Ae^x$, where $A = \left(\frac{c_2+c_3}{c_1}\right)e^{c_4}$

Order = Number of independent arbitrary constants

$= 1$

14. (b)

$x^2dy - 2xydx = x^4 \cos x\, dx$

$\Rightarrow \frac{dy}{dx} = \frac{x^4 \cos x + 2xy}{x^2}$

$\Rightarrow \frac{dy}{dx} - \frac{2}{x}y = x^2 \cos x$

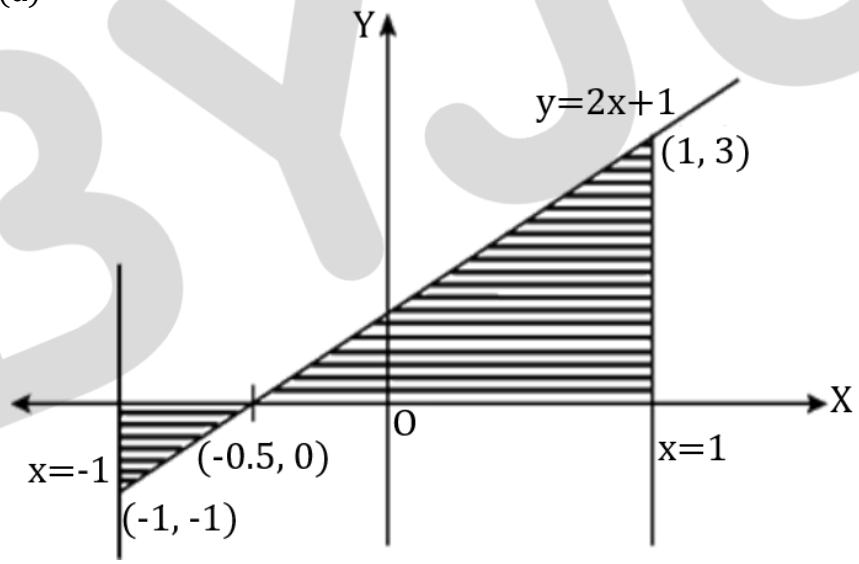
I.F. $= e^{\int -\frac{2}{x}dx} = e^{-2\log x} = \frac{1}{x^2}$

Therefore, the general solution is

$y\left(\frac{1}{x^2}\right) = \int \frac{1}{x^2}(x^2 \cos x)dx = \sin x + c$

$\therefore y = x^2(\sin x + c) = x^2 \sin x + cx^2$

15. (d)

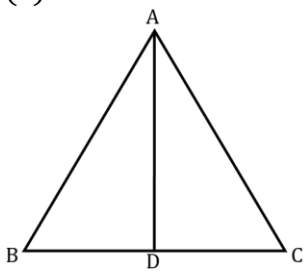


Area bounded by $y = 2x + 1$ with $x-$ axis

$= \frac{1}{2}\left(\frac{1}{2}\right)(1) + \frac{1}{2}\left(\frac{3}{2}\right)(3) = \frac{5}{2}$ sq. units

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16. (a)



$\overrightarrow{AB} = (1, 1, 1)$, $\overrightarrow{AC} = (1, 3, 5)$

$\Rightarrow \overrightarrow{BC} = \overrightarrow{AC} - \overrightarrow{AB} = (0, 2, 4)$

Since D is the mid-point of $\overrightarrow{BC}$, $\therefore \overrightarrow{BD} = (0, 1, 2)$

$\overrightarrow{AD} = \overrightarrow{AB} + \overrightarrow{BD}$

$= (1, 2, 3)$

$\therefore \left|\overrightarrow{AD}\right| = \sqrt{1 + 4 + 9}$

$= \sqrt{14}$

17. (d)

$\left|\vec{a} - \vec{b}\right|^2 = |\vec{a}|^2 + \left|\vec{b}\right|^2 - 2|\vec{a}|\left|\vec{b}\right|\cos\theta$

$= 2(1 - \cos\theta) \quad (\because |\vec{a}| = \left|\vec{b}\right| = 1)$

$\Rightarrow \left|\vec{a} - \vec{b}\right|^2 = 2\left(2\sin^2\frac{\theta}{2}\right)$

$\therefore \sin\frac{\theta}{2} = \frac{|\vec{a} - \vec{b}|}{2}$

18. (a)

Given, $\frac{dy}{dx} = \frac{2x}{y}$

$\Rightarrow ydy = 2xdx$

$\Rightarrow \int ydy = \int 2x\, dx$

$\Rightarrow \frac{y^2}{2} = x^2 + A$, where A is a constant.

The above equation represents a hyperbola.

19. (d)

$\left|\vec{a} \times \vec{b}\right|^2 + \left|\vec{a} \cdot \vec{b}\right|^2 = 144$

$\Rightarrow |\vec{a}|^2\left|\vec{b}\right|^2 \sin^2\theta + |\vec{a}|^2\left|\vec{b}\right|^2 \cos^2\theta = 144$

$\Rightarrow |\vec{a}|^2\left|\vec{b}\right|^2 = 144$

$\Rightarrow \left|\vec{b}\right|^2 = \frac{144}{36} = 4 \quad (\because |\vec{a}| = 6)$

$\Rightarrow \left|\vec{b}\right| = 2$

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20. (d)

Signs of x-coordinate, y-coordinate and z-coordinate are $+,\ -,\ +$ respectively.

$\therefore (1, -3, 4)$ lies in the fourth octant.

21. (b)

Given vectors are coplanar.

$$\Rightarrow \begin{vmatrix} 2 & -3 & 4 \\ 2 & 1 & -1 \\ \lambda & -1 & 2 \end{vmatrix} = 0$$

$$\Rightarrow 2(1) + 3(4 + \lambda) + 4(-2 - \lambda) = 0$$

$$\Rightarrow 2 + 12 + 3\lambda - 8 - 4\lambda = 0$$

$$\Rightarrow \lambda = 6$$

22. (c)

Let $\frac{x-3}{2} = \frac{y-3}{3} = \frac{z+5}{6} = t$

$\Rightarrow (x, y, z) = (2t + 3, 3t + 3, 6t - 5)$

$\therefore$ d.r.'s of the line perpendicular to $\frac{x-3}{2} = \frac{y-3}{3} = \frac{z+5}{6}$ and joining $(2t + 3, 3t + 3, 6t - 5)$ and $(1, 2, -4)$ is $(2t + 2, 3t + 1, 6t - 1)$

$\therefore 2(2t + 2) + 3(3t + 1) + 6(6t - 1) = 0$

$\Rightarrow t = -\frac{1}{49}$

$\therefore$ Distance $= \sqrt{(2t + 2)^2 + (3t + 1)^2 + (6t - 1)^2}$

$$= \sqrt{49t^2 + 2t + 6}$$

$$= \sqrt{\frac{1}{49} - \frac{2}{49} + 6} = \frac{\sqrt{293}}{7}$$

23. (a)

Given line is $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$

and plane is $2x - 2y + z = 5$

So, $\sin\theta = \left|\frac{\vec{b}\cdot\vec{n}}{|\vec{b}||\vec{n}|}\right|$

$$\Rightarrow \sin\theta = \left|\frac{(3\hat{i}+4\hat{j}+5\hat{k})\cdot(2\hat{i}-2\hat{j}+\hat{k})}{\sqrt{9+16+25}\sqrt{4+4+1}}\right|$$

$$= \frac{6-8+5}{\sqrt{50}\sqrt{9}} = \frac{1}{5\sqrt{2}} = \frac{1}{5\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{2}}{10}$$

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24. (b)

Given, $\alpha = \beta = \frac{\pi}{3}$

Let acute angle made by $z-$ axis be γ.

Then, $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

$\Rightarrow \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 + \cos^2\gamma = 1$

$\Rightarrow \frac{1}{4} + \frac{1}{4} + \cos^2\gamma = 1 \Rightarrow \cos^2\gamma = \frac{1}{2}$

$\Rightarrow \cos\gamma = \pm\frac{1}{\sqrt{2}} \Rightarrow \gamma = \frac{\pi}{4}$ [$\because \gamma$ is acute]

25. (c)

Given corner points are $(0, 3), (1, 1), (3, 0)$

$z = px + qy$

At $(3, 0), z = 3p$

At $(1, 1), z = p + q$

It is given that the minimum of z occurs at $(3, 0)$ and $(1, 1)$

$\Rightarrow 3p = p + q \Rightarrow 2p = q \Rightarrow p = \frac{q}{2}$

26. (a)

y-intercept of $x + y = 5$ is $(0, 5)$

y-intercept of $x + 3y = 9$ is $(0, 3)$

The intersection point of $x + y = 5$ and $x + 3y = 9$ is $(3, 2)$

Therefore, the corner points are $(0, 5), (0, 3), (3, 2)$

At $(0, 5), Z = 35$

At $(0, 3), Z = 21$

At $(3, 2), Z = 47$

So, $Z_{\max} = 47$ at $(3, 2)$

27. (c)

Given $n = 10$

Probability of odd number, $p = \frac{1}{2}$

$\therefore q = \frac{1}{2}$

Required probability $= P(X \geq 1)$

$$= 1 - P(X = 0)$$

$$= 1 - {}^{10}C_0 \left(\frac{1}{2}\right)^{10-0} \left(\frac{1}{2}\right)^0$$

$$= 1 - \frac{1}{2^{10}}$$

$$= 1 - \frac{1}{1024} = \frac{1023}{1024}$$

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28. (b)

Given $P(A) = \frac{1}{3}, P(B) = \frac{1}{2}, P(A \cap B) = \frac{1}{6}$

So, $P(A'|B) = 1 - P(A|B)$

$= 1 - \frac{P(A \cap B)}{P(B)}$

$= 1 - \frac{1}{3} = \frac{2}{3}$

29. (b)

Let $P(A|E_1) = x$

By Bayes' theorem,

$P(E_2|A) = \frac{P(E_2)\,P(A \mid E_2)}{P(E_1)\,P(A \mid E_1) + P(E_2)\,P(A \mid E_2)}$

$\Rightarrow \frac{1}{2} = \frac{\left(\frac{1}{2}\right)\left(\frac{2}{3}\right)}{\left(\frac{1}{2}\right)x + \left(\frac{1}{2}\right)\left(\frac{2}{3}\right)}$

$\Rightarrow x = \frac{2}{3}$

$\therefore P(E_1|A) = \frac{P(E_1)\,P(A|E_1)}{P(E_1)P(A|E_1) + P(E_2)\,P(A|E_2)}$

$= \frac{\left(\frac{1}{2}\right)\left(\frac{2}{3}\right)}{\left(\frac{1}{2}\right)\left(\frac{2}{3}\right) + \left(\frac{1}{2}\right)\left(\frac{2}{3}\right)} = \frac{1}{2}$

30. (c)

Required probability $= P(A'BC) + P(AB'C) + P(ABC')$

$= \frac{1}{2} \times \frac{1}{4} \times \frac{1}{3} + \frac{1}{2} \times \frac{3}{4} \times \frac{1}{3} + \frac{1}{2} \times \frac{1}{4} \times \frac{2}{3}$

$= \frac{1}{24} + \frac{1}{8} + \frac{1}{12}$

$= \frac{1+3+2}{24} = \frac{1}{4}$

31. (a)

$n(A) = 2$

Given, $2^{n(A) \cdot n(B)} = 1024$

$\Rightarrow (2)^{2 \cdot n(B)} = (2)^{10}$

$\Rightarrow 2 \cdot n(B) = 10$

$\Rightarrow n(B) = 5$

32. (b)

$\sin^2 51° + \sin^2 39°$

$= \cos^2 39° + \sin^2 39°$

$= 1$

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33. (b)

$\tan A + \cot A = 2$

$\Rightarrow (\tan A + \cot A)^2 = 4$

$\Rightarrow \tan^2 A + \cot^2 A + 2\tan A \cot A = 4$

$\Rightarrow \tan^2 A + \cot^2 A = 2$

$\Rightarrow (\tan^2 A + \cot^2 A)^2 = 4$

$\Rightarrow \tan^4 A + \cot^4 A + 2\tan^2 A \cot^2 A = 4$

$\Rightarrow \tan^4 A + \cot^4 A = 2$

34. (d)

Total number of subsets of A is $2^{n(A)} = 2^6 = 64$

Number of subsets of A which contain at least two elements is

$64 - ({}^6C_0 + {}^6C_1)$

$= 64 - (1 + 6)$

$= 57$

35. (a)

$|z + 1| = |z - 1|$

Let $z = x + iy$

Then, $|x + iy + 1| = |x + iy - 1|$

$\Rightarrow \sqrt{(x+1)^2 + y^2} = \sqrt{(x-1)^2 + y^2}$

$\Rightarrow (x+1)^2 + y^2 = (x-1)^2 + y^2$

$\Rightarrow x^2 + 2x + 1 + y^2 = x^2 + 1 - 2x + y^2$

$\Rightarrow 4x = 0$

$\Rightarrow x = 0$ which represents $y -$ axis.

36. (b)

${}^{16}C_9 + {}^{16}C_{10} - {}^{16}C_6 - {}^{16}C_7$

$= {}^{16}C_9 + {}^{16}C_{10} - {}^{16}C_{10} - {}^{16}C_9 = 0 \qquad (\because {}^nC_r = {}^nC_{n-r})$

37. (b)

Number of terms in the expansion of $(x + y + z)^{10}$ is

${}^{10+3-1}C_{10}$

$= {}^{12}C_{10} = \frac{12!}{2!\,10!} = 66$

38. (d)

For $n = 1, 2, 3, 2^n > n!$

$P(4): 2^4 < 4!$

So, smallest positive integer, $n = 4$

39. (b)

Product of slopes $= -1 \Rightarrow ll' + mm' = 0$

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40. (c)

$x^2 = 4ay$

Given parabola passes through $(2, 1)$.

$\Rightarrow 2^2 = 4a \Rightarrow a = 1$

Length of latus rectum $= 4a = 4$

41. (d)

$S_n = n^2 + n$

$S_1 = 1 + 1 = 2 = T_1$

$S_2 = 2^2 + 2 = 6 = T_1 + T_2$

$\therefore T_2 = S_2 - S_1 = 4$

Common difference, $d = T_2 - T_1 = 4 - 2 = 2$

42. (d)

Negation: For some real numbers x and y, $x + y \neq y + x$

43. (b)

Mean, $\overline{x} = \frac{6+7+8+9+10}{5} = \frac{40}{5} = 8$

Standard deviation, $\sigma = \sqrt{\frac{1}{5}(4 + 1 + 0 + 1 + 4)}$ $\left(\because \sigma = \sqrt{\frac{1}{n}\sum(x_i - \overline{x})^2}\right)$

$\Rightarrow \sigma = \sqrt{\frac{10}{5}} = \sqrt{2}$

44. (b)

$\lim_{x \to 0}\left(\frac{\tan x}{\sqrt{2x+4} - 2}\right)$ $\left[\frac{0}{0}\right]$ form

Applying L' Hopitals' rule, we get

$\lim_{x \to 0}\left(\frac{\sec^2 x}{\frac{2}{2 \times \sqrt{2x+4}} - 0}\right)$

$= \lim_{x \to 0}\left(\left(\sqrt{2x+4}\right)\sec^2 x\right)$

$= \sqrt{2 \times 0 + 4} \times 1 = 2$

45. (d)

$R = \{(1, 1)\}$ on set $\{1, 2, 3\}$

Clearly, R is symmetric and transitive.

46. (c)

$f(x) = (x - 2)^2 + 1 \geq 1, \forall\, x \in [2, \infty)$

$f_{\min} = 1$ at $x = 2$

$\therefore$ Range of f is $[1, \infty)$

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47. (d)

Given, $P(A) = 2P(B) = 3P(C)$

$\Rightarrow P(C) = \frac{2}{3}P(B)$

Since A, B, C are three mutually exclusive and exhaustive events

$\therefore P(A) + P(B) + P(C) = 1$

$\Rightarrow P(B) = \frac{3}{11}$

48. (b)

For f to be defined,

$x - 1 \geq 0$ and $-1 \leq \sqrt{x-1} \leq 1$

$\Rightarrow x \geq 1$ and $0 \leq x - 1 \leq 1$

$\Rightarrow x \geq 1$ and $1 \leq x \leq 2$

$\Rightarrow x \in [1, 2]$

Hence, domain of f is $[1, 2]$.

49. (a)

$\frac{\pi}{3} \notin [-1, 1]$ which is the domain of $\sin^{-1} x$, $\cos^{-1} x$

So, $\cos\left(\sin^{-1}\frac{\pi}{3} + \cos^{-1}\frac{\pi}{3}\right)$ does not exist.

50. (d)

$$A^2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} = I$$

$\therefore A^4 = (A^2)^2 = I^2 = I$

51. (a)

$A = \{a, b, c\}$

$n(A) = 3$

Number of binary operations is $n(A)^{(n(A))^2} = 3^{3^2} = 3^9$

52. (c)

We know that if $BA = AB = I$, then A and B are inverse of each other.

Let $B = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix} \Rightarrow |B| = 1$

$\text{adj}(B) = \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}$

$B^{-1} = \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}$

$\therefore \ A = B^{-1} = \begin{pmatrix} 2 & -1 \\ -3 & 2 \end{pmatrix}$

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53. (d)

$$f(0) = \begin{vmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{vmatrix}$$

$f(0)$ is the determinant of skew-symmetric matrix of order 3 (odd).

$\therefore f(0) = 0$

54. (a)

B is a skew symmetric matrix.

$\Rightarrow B' = -B$

Now, $(A'BA)' = A'B'(A')'$

$= A'(-B)A$

$= -(A'BA)$

Hence, $A'BA$ is a skew symmetric matrix.

55. (c)

$|A \text{ adj } A| = |A||\text{adj } A|$

$= |A||A|^{3-1}$

$= 5 \times 5^2 = 125$

56. (a)

Given, f is continuous at $x = 0$

$\Rightarrow \lim\limits_{x \to 0} f(x) = f(0)$

$\Rightarrow \lim\limits_{x \to 0} \frac{1-\cos Kx}{x \sin x} = \frac{1}{2}$

Applying L' Hopitals' rule, we get

$\lim\limits_{x \to 0} \frac{K \sin Kx}{\sin x + x \cos x} = \frac{1}{2}$

Again, by L' Hopitals' rule,

$\lim\limits_{x \to 0} \left(\frac{K^2 \cos Kx}{\cos x - x \sin x + \cos x}\right) = \frac{1}{2}$

$\Rightarrow \frac{K^2}{1-0+1} = \frac{1}{2}$

$\Rightarrow K = \pm 1$

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57. (d)

Let d be the common difference of A.P.

Then, $\Delta = \begin{vmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{vmatrix} = \begin{vmatrix} a_1 & a_1+d & a_1+2d \\ a_1+3d & a_1+4d & a_1+5d \\ a_1+6d & a_1+7d & a_1+8d \end{vmatrix}$

Applying $C_2 \to C_2 - C_1$ and $C_3 \to C_3 - C_2$

$\Delta = \begin{vmatrix} a_1 & d & d \\ a_1+3d & d & d \\ a_1+6d & d & d \end{vmatrix}$

$= 0$ ($\because C_2$ and C_3 are identical)

$= \log_e(\log_e e)$ ($\because \log_e 1 = 0$)

58. (c)

$2^x + 2^y = 2^{x+y}$ $\cdots (1)$

Differentiating both sides w.r.t. x, we get

$2^x \ln 2 + 2^y \ln 2 \cdot y' = 2^{x+y} \ln 2\,(1 + y')$

$\Rightarrow 2^x + 2^y \cdot y' = 2^{x+y}(1 + y')$

$\Rightarrow 2^x + 2^y \cdot y' = 2^{x+y} + 2^{x+y} \cdot y'$

$\Rightarrow 2^x - 2^{x+y} = y'(2^{x+y} - 2^y)$

$\Rightarrow 2^x - 2^x - 2^y = y'(2^x + 2^y - 2^y)$ [From eqn.(1)]

$\Rightarrow -2^y = y' 2^x$

$\Rightarrow y' = \frac{dy}{dx} = -2^{y-x}$

59. (c)

$f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$

Put $x = \tan\theta$, where $\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$f(x) = \sin^{-1}\left(\frac{2\tan\theta}{1+\tan^2\theta}\right)$

$\Rightarrow f(x) = \sin^{-1}(\sin 2\theta)$

$\Rightarrow f(x) = 2\theta = 2\tan^{-1}x$ ($\because \theta = \tan^{-1}x$)

$f'(x) = \frac{2}{1+x^2}$

$\therefore f'(\sqrt{3}) = \frac{2}{1+3} = \frac{1}{2}$

60. (c)

LHL $= \lim_{x\to 0^-}\left(\frac{e^{1/x}-1}{e^{1/x}+1}\right) = \frac{e^{-\infty}-1}{e^{-\infty}+1} = \frac{0-1}{0+1} = -1$

Dividing both numerator and denominator by $e^{1/x}$, we get

$\lim_{x\to 0}\left(\frac{1-e^{-1/x}}{1+e^{-1/x}}\right)$

RHL $= \lim_{x\to 0^+}\left(\frac{1-e^{-1/x}}{1+e^{-1/x}}\right) = \frac{1-e^{-\infty}}{1+e^{-\infty}} = \frac{1-0}{1+0} = 1$

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